Eco-Track Smart Waste Management System

Mini Project – Week 1

Prepared By:
Sahana Manjunath Raikar

Role: System Architecture & Data Flow

Introduction

Urban areas often face serious challenges in waste management due to inefficient collection systems and lack of real-time monitoring. Overflowing garbage bins and illegal dumping not only create unpleasant surroundings but also lead to environmental pollution and health risks.

Traditional waste management systems rely on fixed schedules for waste collection, which means bins may overflow before they are emptied. Additionally, citizens often do not have a convenient way to report waste issues to municipal authorities.

The **Eco-Track Smart Waste Management System** aims to solve these problems by using smart technology. Smart bins equipped with sensors monitor the fill level of garbage bins and send real-time data to a central system. Citizens can also report overflowing bins or illegal dumping through a mobile application.

The system collects all this information and displays it on a centralized dashboard used by municipal authorities. It also includes a **route optimization module** that suggests the most efficient routes for waste collection vehicles. This approach improves efficiency, reduces operational costs, and helps maintain cleaner cities.

System Overview

The Eco-Track Smart Waste Management System is a digital platform designed to improve waste collection efficiency using smart monitoring and citizen participation.

The system integrates three main elements:

- Smart bin sensors that monitor bin fill levels
- A citizen mobile application for reporting waste issues
- An admin dashboard for municipal monitoring and decision-making

When a garbage bin becomes full, the smart sensor detects the fill level and sends the data to the backend server. Citizens can also report waste problems through the mobile application by uploading images and location details.

The backend server processes the data and stores it in a database. Municipal authorities can view this information through the admin dashboard and take necessary action. The system also analyzes bin status data and generates optimized routes for waste collection vehicles.

This smart approach ensures that waste is collected efficiently and helps prevent overflowing bins.

System Components

The Eco-Track system consists of several key components that work together.

1. Citizen Mobile Application

The mobile application allows citizens to report waste management issues such as overflowing bins or illegal dumping. Users can upload photos, provide descriptions, and share their location.

2. Smart Bin Sensors

Smart sensors installed in garbage bins detect the fill level of waste. When the bin reaches a certain level, the sensor automatically sends the information to the backend server.

3. Backend Server

The backend server acts as the central processing system. It receives data from mobile applications and smart bin sensors, processes the information, and stores it in the database.

4. Database

The database stores important system information such as citizen reports, sensor data, bin locations, and waste collection schedules.

5. Admin Dashboard

The admin dashboard is used by municipal authorities to monitor waste management activities. It displays real-time bin status, citizen reports, and route optimization results.

6. Route Optimization Module

This module analyzes bin fill levels and determines the most efficient routes for waste collection trucks.

System Architecture

System architecture describes the structure of the system and how different components interact with each other.

The Eco-Track system architecture consists of three major layers:

- User Layer
- Application Layer
- Data Layer

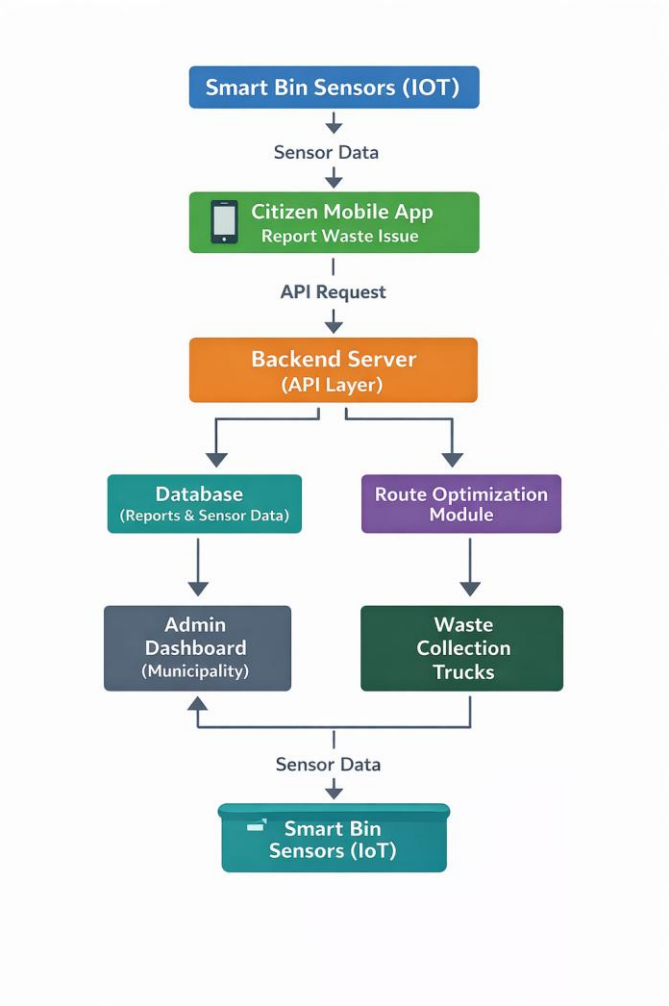
The **User Layer** includes citizens using the mobile application and municipal authorities using the admin dashboard.

The **Application Layer** consists of the backend server that processes incoming requests and handles system operations.

The **Data Layer** stores all system data including reports and sensor readings.

System Architecture Diagram

This diagram shows how different system components communicate with each other. Citizen reports and sensor data are processed by the backend server, stored in the database, and displayed on the admin dashboard.



Data Flow

Data flow represents how information moves between different components in the system.

The Eco-Track system collects data from two main sources:

- Citizen reports
- Smart bin sensors

Both types of data are processed by the backend server and stored in the central database.

Citizen Report Data Flow

1. A citizen reports a waste issue using the mobile application.

2. The mobile application sends the report to the backend server through an API request.
3. The backend server processes the request and stores the report in the database.
4. The admin dashboard retrieves the data from the database.
5. Municipal authorities view the report and take action.

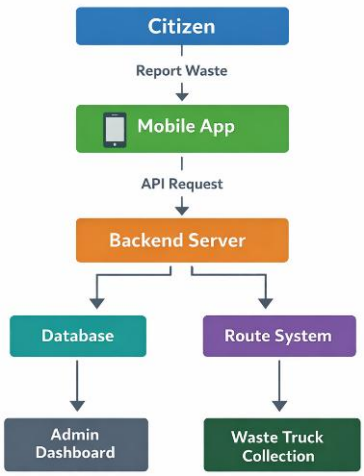
Smart Bin Sensor Data Flow

1. Smart bin sensors monitor the fill level of waste bins.
2. When the bin reaches a certain threshold, the sensor sends data to the backend server.
3. The backend server updates the bin status in the database.
4. The admin dashboard displays the updated bin status.

Route Optimization Data Flow

1. The system collects bin status data from the database.
2. The route optimization module analyzes which bins need collection.
3. The module generates an optimized waste collection route.
4. The route is displayed on the admin dashboard for the waste collection team.

Data Flow Diagram (DFD)



Benefits of the System

The Eco-Track Smart Waste Management System provides several benefits:

- Real-time monitoring of waste bin levels
- Faster reporting of waste issues by citizens
- Improved communication between citizens and municipal authorities
- Optimized waste collection routes
- Reduced environmental pollution caused by overflowing bins
- Efficient resource management for waste collection teams

Conclusion

The Eco-Track Smart Waste Management System offers a modern solution for improving urban waste management. By integrating smart bin sensors, citizen reporting, and route optimization, the system provides real-time monitoring and efficient waste collection.

The platform enables municipal authorities to respond quickly to waste issues and maintain cleaner environments. With the help of smart technology and data-driven decision making, the Eco-Track system contributes to building sustainable and smart cities.